



Echocardiography

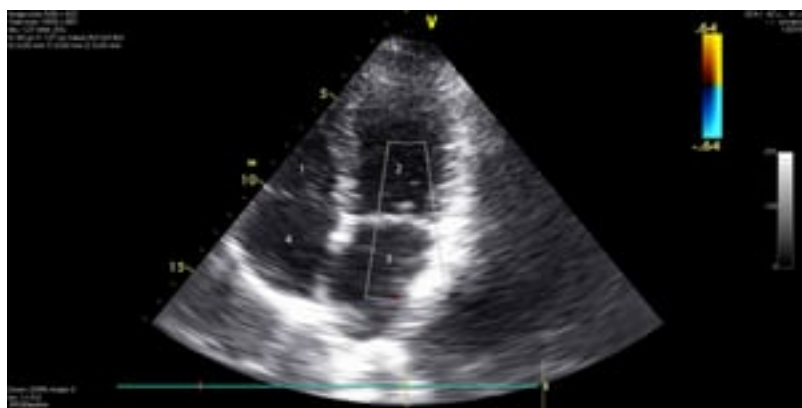
Full Review: May 2026 By [Thomas Cascino](#), MD, MSc, Michigan Medicine, University of Michigan | [Michael J. Shea](#), MD, Michigan Medicine at the University of Michigan | Peer reviewed by [Jonathan G. Howlett](#), MD, Cumming School of Medicine, University of Calgary
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Echocardiography uses ultrasound waves to produce an image of the heart, the heart valves, and the great vessels. It also uses the Doppler principle to create color reconstruction (color Doppler) and waveform images (spectral Doppler) of blood flow. Echocardiography can assess:

- Cardiac and great vessel anatomy
- Systolic function as well as diastolic filling patterns of the ventricles (useful in left ventricular hypertrophy, [hypertrophic](#) or [restrictive cardiomyopathy](#), severe [heart failure](#), and [constrictive pericarditis](#))
- Wall motion (providing information about ischemia and infarction)
- Structure and function of the heart valves
- Heart wall thickness (eg, in hypertrophy or atrophy)
- Valvular vegetations and intracardiac mass/thrombus
- Pulmonary artery and central venous pressure estimates

Echocardiography is useful in assessing a variety of cardiac conditions and is widely available, safe, and relatively inexpensive. For most cardiac conditions, it remains the initial imaging modality of choice, although it is not always the definitive or ultimate diagnostic modality.

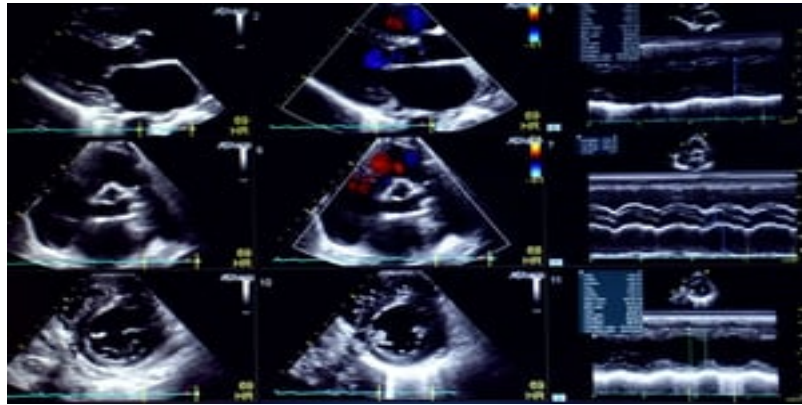
Echocardiographic Views and Techniques



Echocardiogram (4-Chamber View)

This image shows all 4 cardiac chambers and the tricuspid and mitral valves. 1 = right ventricle; 2 = left ventricle; 3 = left atrium; 4 = right atrium.

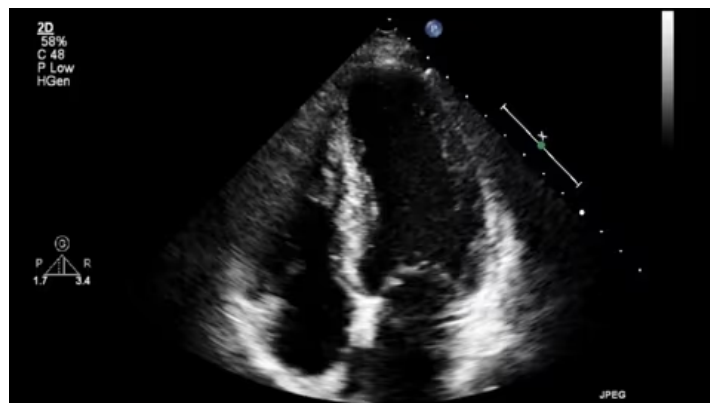
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Echocardiogram (9-Panel Plot)

This 9-panel plot shows additional echocardiographic views and techniques, including a parasternal long axis view (upper left image), parasternal short axis views (middle left, lower left, lower middle images), color Doppler images (upper middle and central images), and M-mode (right upper, middle, and lower images).

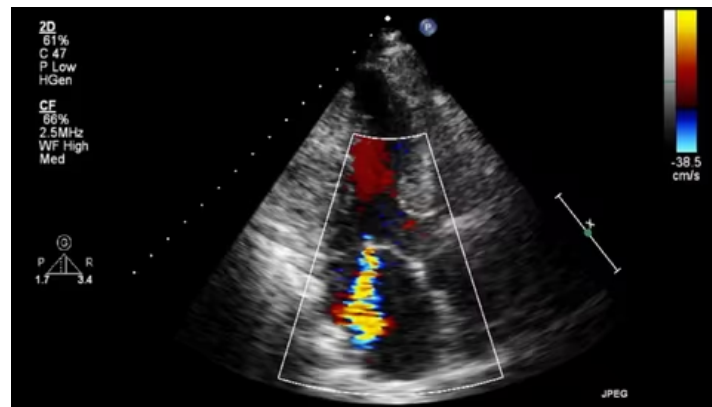
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Transthoracic Echocardiogram

This transthoracic echocardiogram shows a 4-chamber view of the heart: right atrium (lower left), right ventricle (upper left), left atrium (lower right), and left ventricle (upper right).

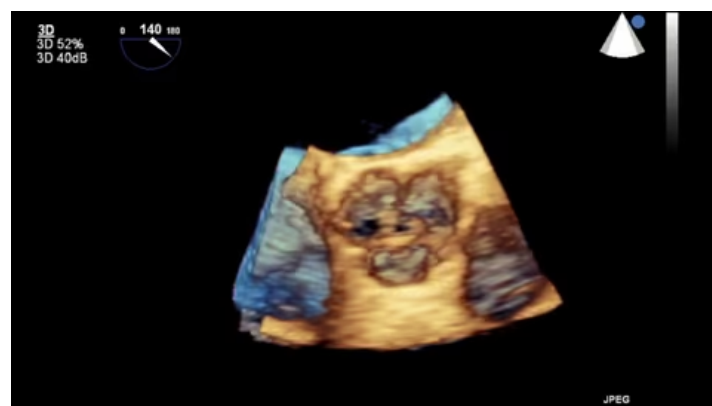
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Transthoracic Color Doppler Echocardiogram

This transthoracic echocardiogram uses color Doppler imaging to show blood flowing into the left ventricle across the mitral valve (red flow across the valve, center left), blood flowing out of the aortic valve (yellow-blue flow diagonally across the valve, center right), and blood leaking backwards across the mitral valve (yellow-blue flow pointed towards the bottom). Note the right and left orientation is reversed when compared with the previous video.

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Transesophageal 3D Echocardiogram

This transesophageal 3D echocardiogram shows the aortic valve opening and closing.

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Techniques

There are 3 primary techniques for performing echocardiography:

- Transthoracic

- Transesophageal
- Intracardiac

Transthoracic echocardiography (TTE) is the most common echocardiographic technique. In TTE, the transducer is used to obtain various views from the left (or right) sternal border, at the cardiac apex, at the suprasternal notch (to allow visualization of the aortic valve, left ventricular outflow tract, and descending aorta), and over the subxiphoid region. TTE provides 2-dimensional (and sometimes 3-dimensional) still, cinematic, and Doppler images of most major cardiac structures. TTE is a relatively inexpensive and non-invasive imaging technique for diagnosis of right and left ventricular function and wall motion, chamber size and anatomy, valvular structure and function, aortic root structure, and intracardiac pressures estimates.

Point-of-care ultrasound (POCUS) is a limited TTE (focused on detecting significant pericardial effusion and ventricular dysfunction) that is sometimes performed at the bedside of critically ill patients in the intensive care unit (ICU) and emergency department (ED); many intensivists and emergency department physicians have training to do this procedure with portable hand-held machines to detect conditions requiring emergent treatment, or when experienced radiologists or cardiologists are not available ([1](#), [2](#)). Portable hand-held machines act as a screening tool to determine which patients may require more detailed testing. With expanding use by less experienced clinicians, the major limitation is missed diagnoses. As a result, national societies are making recommendations for training in cardiovascular POCUS to facilitate the best use of the diagnostic test. Sites using cardiovascular POCUS should develop standards for use within their practice.

In **transesophageal echocardiography** (TEE), a transducer on the tip of an endoscope passed into the esophagus and stomach allows visualization of the heart. TEE is used to assess cardiac disorders when transthoracic study is technically difficult, as in patients with obesity or chronic obstructive pulmonary disease (COPD). It reveals better detail of small abnormal structures (eg, endocarditic vegetations or patent foramen ovale) and posterior cardiac structures (eg, left atrium, left atrial appendage, interatrial septum, pulmonary vein anatomy) because they are closer to the esophagus than to the anterior chest wall. TEE can also produce images of the ascending aorta, which arises behind the third costal cartilage; of structures < 3 mm (eg, thrombi, vegetations); and of prosthetic valves. TEE also plays an important role in the cardiac operating room and cardiac catheterization laboratory, allowing pre- and post-procedure assessment as well as intraprocedural guidance.

In **intracardiac echocardiography** (ICE), a transducer on the tip of a catheter (inserted via the femoral vein and threaded to the heart) allows visualization of cardiac anatomy. ICE can be performed during structural cardiac (eg, percutaneous closure of atrial septal defects or patent foramen ovale) or electrophysiologic procedures. ICE provides better image quality and decreased procedure time when compared with TEE during many procedures ([3](#)). However, ICE is generally more expensive ([4](#)).

Epicardial echocardiography is a technique used in the operating room when the chest is open, for which the transducer is placed directly on the epicardium to obtain limited images.

Methodology

Two-dimensional echocardiography is the basis for echocardiographic studies. A transducer containing piezoelectric crystals transmits and receives reflected sound waves. The data from the transducer is used to reconstruct a 2-dimensional still or moving image of the heart and surrounding structures, from which anatomic and functional information is ascertained.

Spectral Doppler echocardiography uses the Doppler principle to display the velocity and direction of blood flow as a waveform. This technique is useful for detecting abnormal blood flow (eg, due to regurgitant lesions) or velocity (eg, due to stenotic lesions). Spectral Doppler echocardiography does not provide spatial information about the size or shape of the heart or its structures.

Color Doppler echocardiography combines 2-dimensional and spectral Doppler echocardiography to provide information about the size and shape of the heart and its structures as well as the velocity of and direction of blood flow around the valves and outflow tracts. Color is used to code blood flow information; by convention, red is toward and blue away from the transducer.

Contrast echocardiography is a 2-dimensional echocardiogram performed while agitated saline or another ultrasonographic contrast agent is rapidly injected into the cardiac circulation. Agitated saline develops microbubbles, which produce a cloud of echoes in the right cardiac chambers and which appear on the left side of the heart only if a right-to-left shunt such as a septal defect is present. Other contrast agents readily traverse the pulmonary capillary bed and can therefore be used to delineate the heart chambers, especially the left ventricle, in patients with poor transthoracic echocardiographic windows.

Tissue Doppler imaging uses Doppler techniques to measure the velocity of myocardial tissue contraction (rather than of blood flow). Myocardial tissue movement can also be evaluated with **speckle-tracking echocardiography**, which uses algorithms to track myocardial echo speckles (characteristic reverberations from the myocardium during an ultrasound) from frame to frame. **Strain imaging** uses these data to calculate myocardial strain (percentage change in length between contraction and relaxation) and myocardial strain rate (rate of change in length). Strain and strain rate measurements can help assess systolic and diastolic function and identify ischemia during stress testing.

In **3-dimensional echocardiography**, special transducers contain an array of crystals; the data received are used to reconstruct a 3-dimensional image of cardiac structures. Three-dimensional echocardiography is particularly useful in evaluating the mitral valve apparatus (or other valves) for surgical correction. It is also used to quantify ventricular volumes, including that of the right ventricle, because the crescent shape of the right ventricle limits evaluation by 2-dimensional echocardiography, and during interventional procedures such as transcatheter edge-to-edge repair of the mitral valve.

Stress echocardiography

Transthoracic echocardiography is an alternative to [radionuclide imaging](#) and cardiac MRI to identify areas of myocardial ischemia during and after exercise or pharmacologic [stress testing](#). Stress echocardiography shows regional wall motion abnormalities that result from an imbalance in blood flow in epicardial coronary vessels during stress. Computer programs can provide side-by-side assessment

of ventricular contraction during systole and diastole at rest and under stress. Exercise and pharmacologic protocols are similar to those used in radionuclide stress testing. Stress echocardiography is approximately equivalent to radionuclide stress testing and for detecting ischemia, although stress cardiac MRI has most diagnostic accuracy (5). The choice between tests is often based on availability, the provider's experience, and cost.

Stress echocardiography is valuable in evaluating the hemodynamic severity of aortic valve stenosis in patients with significant symptoms but whose resting transvalvular pressure gradient is not markedly high.

Evidence of high left ventricular filling pressures during exercise, including elevated E/e' ratio (ratio of transmitral flow to mitral annular velocity) or elevated tricuspid regurgitant velocity, can be used to diagnose [heart failure with preserved ejection fraction](#) or dynamic valvular disease such as mitral regurgitation.

References

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5. [Bax JJ, Di Carli M, Narula J, Delgado V.](#) Multimodality imaging in ischaemic heart failure. *Lancet.* 2019;393(10175):1056-1070. doi:10.1016/S0140-6736(18)33207-0



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